

Cloud Architecture

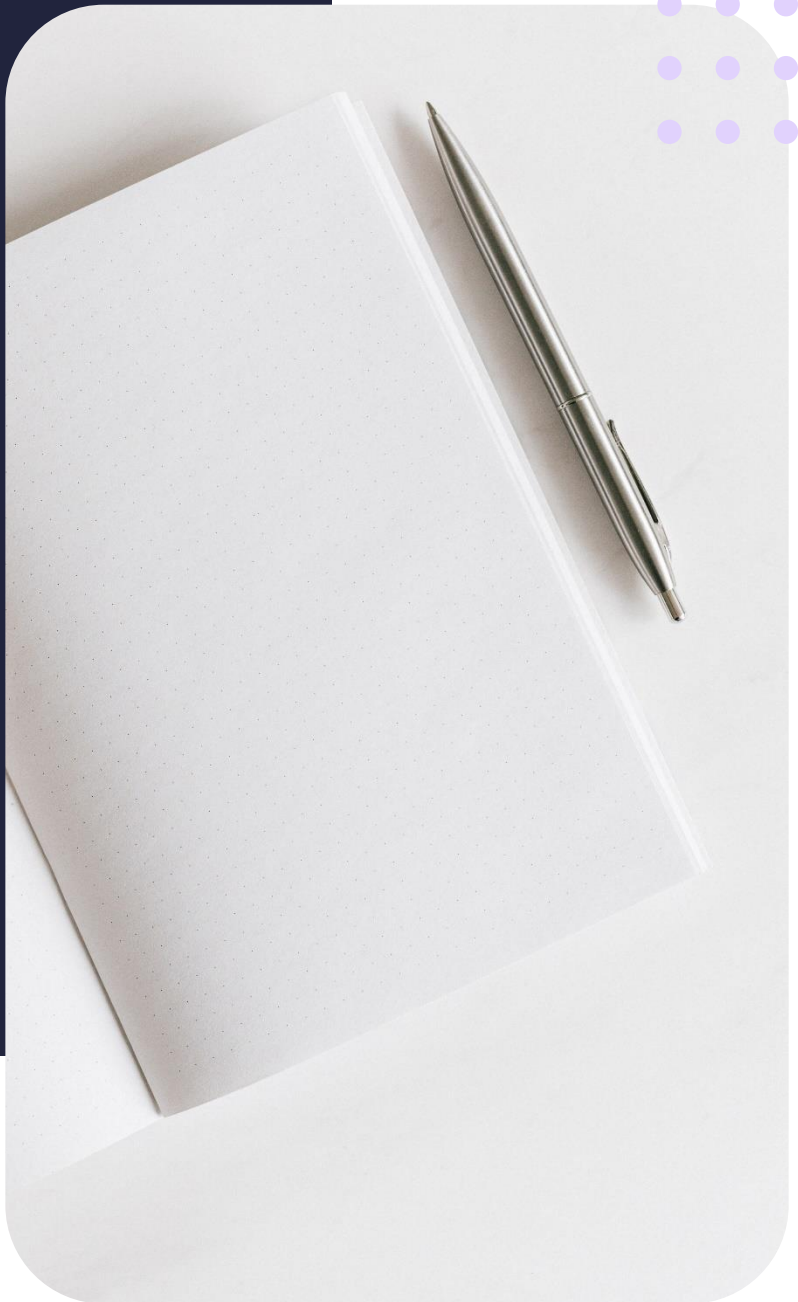
Introduction to cloud architecture

Summary Notes



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Lesson objectives

By the end of this lesson, you should be able to:

- Define cloud computing
- Describe cloud computing architectures
- Explore applications

Traditional computing

Traditional computing makes use of hardware and software within a single central location. Traditional computing requires organisations to procure hardware, software, and other services as well as staff to manage and maintain the computing resources. This type of computing can be expensive for organisations to implement and maintain. Furthermore, traditional computing does not guarantee 100% uptime and may be prone to security issues.

Cloud computing

Cloud computing is quite the opposite of traditional computing that allows users to access software applications, hardware, and other services over a shared network such as the internet or a private network. It is regarded as a form of distributed computing that enables users to store data and perform computation on a remote server rather than from a personal server. Cloud computing tends to be modular and flexible in design and is less expensive to maintain than traditional computing.

Advantages of cloud computing

- Cloud computing is a lot more scalable than traditional computing. Software applications and hardware can be extended to meet user demands.
- With cloud computing organisations are likely to experience a higher level of business continuity since data is often backed up and systems have a higher degree of uptime.
- Collaboration is a lot more efficient with the use of cloud computing. This is possible given that multiple individuals are able to access and work on a single platform simultaneously.
- Cloud computing can help businesses reduce operating costs since there is no total cost of ownership where infrastructure and maintained is concerned.

Disadvantages of cloud computing

- Since accessing cloud computing services requires a network, the availability and performance of the network will dictate the user's experience with the cloud.
- While inexpensive in the short-run, the pay-to-scale model adopted by most cloud providers can become expensive for organisations in the long run.
- Given that cloud providers own the underlying infrastructure that powers the cloud, the collapse of a server or data centre could result in the loss of an organisations control over their data.
- Even though security on the cloud is arguably better than traditional computing, security remains a concern since organisations are dependent on the security implemented by cloud providers.

Types of cloud computing services

Infrastructure as a service (IaaS)

The most common type of cloud computing services that allows users to rent IT infrastructure from a cloud provider. With IaaS, the cloud provider manages and maintains the data centres, servers, as well as other physical infrastructure, while providing virtualised instances of the infrastructure to organisations.

Software as a service (SaaS)

SaaS is an approach used for delivering software applications over the internet. This is usually a complete software application that the organisation has access to. Physical infrastructure such as servers and data centres are managed and maintained by the cloud provider.

Platform as a service (PaaS)

PaaS is supply on-demand environment for developing, testing, delivering and managing software applications. PaaS provides an organisation a framework for software development, this is often useful for organisations that specialise in developing software projects.

Applications of cloud computing

Business use cases

- Business intelligence
- Data warehousing
- Data backup

Personal use cases

- Storage
- Emails
- Gaming

Cloud service providers

A cloud provider or cloud service provider is an external company which offers a cloud-based platform to organisations. This could may include services such as IaaS, PaaS and SaaS.

Examples of popular cloud providers

- Microsoft Azure
- Amazon Web Services (AWS)
- IBM Cloud